

HIRDESH SINGH

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PROFESSIONAL SUMMARY

Computer Science undergraduate and full-stack developer with hands-on experience building responsive, production-style web applications using JavaScript/TypeScript, React.js, Node.js, and REST APIs. Comfortable working across the MVC stack and versioncontrolled (Git) environments, with a strong foundation in Object-Oriented Programming, Data Structures & Algorithms, and Database Management Systems (MySQL, MongoDB, and working knowledge of relational databases such as PostgreSQL). Proven interest in AI/ML and workflow automation through applied projects in computer vision and deep learning. Agile team collaborator, self-starter, and quick learner eager to pick up new frameworks such as Ruby on Rails and Flutter.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript (working), Java, SQL, Python, C++

Frontend: React.js, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, REST API Design, MVC Architecture, Spring Boot/MVC (Basic), JDBC (Basic)

Databases: MongoDB, MySQL; working knowledge of PostgreSQL

Tools & Practices: Git, GitHub, Postman, VS Code, Agile/Scrum, CI/CD (familiar), Debugging & Testing

AI/ML & Automation: CNNs, RNNs, LSTMs, YOLOv8, OpenCV — applied to intelligent, agent-style automation projects

Core CS Concepts: OOP, Data Structures & Algorithms, DBMS, Operating Systems, SDLC, RESTful Web Services **PROJECTS**

Medicare – Online Disease Prediction & Doctor Appointment System

HTML, CSS, JavaScript, MongoDB, REST APIs

Live: <https://doctor-meds-frontend.onrender.com>

- Built responsive, multi-role UIs (Admin, Doctor, User) with HTML5, CSS3, and JavaScript for appointment booking and diseaseprediction workflows.
- Designed MongoDB data models and implemented full CRUD operations for users, doctors, and appointments.
- Built and integrated REST APIs for authentication, scheduling, and role-based access control.
- Applied OOP principles and modular design to improve maintainability and scalability — directly transferable to full-stack, agile feature development.

NextGen Ballot – Online Voting Platform

Node.js, Express.js, MongoDB Atlas, Mongoose, HTML, CSS, JavaScript

Live: <https://nextgen-ballot.onrender.com>

- Built a secure full-stack online voting application using Node.js, Express.js, MongoDB Atlas, and Mongoose.
- Implemented OTP authentication, CAPTCHA verification, Aadhaar validation, and session-based login for secure user authentication.
- Developed REST APIs for voter registration, vote casting, result retrieval, and administrative operations.
- Deployed the application on Render with MongoDB Atlas and managed version control using Git and GitHub.
- Designed responsive frontend pages using HTML, CSS, and JavaScript with secure backend integration.

Smart Navigation System for Visually Impaired

Python, YOLOv8, OpenCV, Flask, pyttsx3

- Built a real-time, AI-based assistive automation system using YOLOv8 and OpenCV to detect obstacles from live camera input.
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- Designed a modular processing pipeline (detection → decision-making → voice alerts) integrated with a Flask backend and web interface.
- Collaborated in an Agile workflow to integrate the computer-vision pipeline with the backend and UI, working in sprints and iterating on feedback.

EDUCATION

Bachelor of Technology – Computer Science & Engineering

Aug 2023 – Aug 2027 (Expected)

VIT Bhopal University, Bhopal | CGPA: 8.47/10

Class XII – Sachdeva Millennium School (94.6%) | Class X – Sachdeva Millennium School (94.8%)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Computer Networks, Web Technologies

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Solved 150+ Data Structures & Algorithms problems on LeetCode (arrays, trees, graphs, dynamic programming) using Java.
- Solved 100+ problems on GeeksforGeeks, reinforcing core algorithmic problem-solving.
- Participated in multiple university-level hackathons, collaborating in agile teams to design and ship scalable web solutions under tight deadlines.

CERTIFICATIONS

- Introduction to Machine Learning – NPTEL
- Marketing Analytics – NPTEL
- Oracle Certified Foundations Associate – Oracle
- Generative AI Professional – Oracle
- Google IT Support Professional Certificate – Google / Coursera
- Bits and Bytes of Computer Networking – Coursera
- AWS Certification Course – Intellipaat

SOFT SKILLS

Team Collaboration • Problem Solving • Communication • Adaptability • Time Management • Attention to Detail • Ownership